



Task 1 – Pass Submission

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A. Task Pass1a

<i>Feature</i>	<i>Query</i>	<i>Filter</i>
Primary Purpose	Build a reusable, custom dataset or perform actions to data	Temporarily narrow down records in an existing view.
Permanence	Saved as a permanent object in the database navigation pane.	Temporary, frequently disappears when the form or table is closed.
Multi-Table Support	Can combine and link data from multiple related tables.	Works only on the currently active table, form, or report.
Data Alteration	Can calculate fields, group data, or update/delete records.	Can only hide or show existing records; cannot alter data.
As a Data Source	Can be used as the data source for forms, reports, or other queries.	Cannot be used as a standalone data source for other objects.

Based on the comparison table above, it is clear that the Query method has various advantages that over using filters:

- Due to the permanent nature, once a query is generated and saved, it remains forever until it is modified or latest update while filter is temporary which contrast to Query.
- Secondly, Query supports multi-table connect and support function that helps to combine and link data from multiple related tables into the only single view.
- Finally, regarding of data alteration, Query performs calculations, group data, or update/delete records to create new information on the fly while filter can only hide or show existing records.

After that, I opened the file named *moviedatabaseDAD.accdb* which downloaded from Canvas in the *Datasheet View* according to the requirement.

ActorNo	FullName	GivenName	Surname	Gender	Birthdate	BirthCountry	Password	NoOfDogs	NoOfCats	Click to Add
3	Harrison Ford	Harrison	Ford	M	7/13/1942	USA	AbAaAb	3	3	
7	Andrew Stanton	Andrew	Stanton	M	12/3/1965	USA	aaabaD	1	3	
8	Lee Unkrich	Lee	Unkrich	M	8/8/1967	USA	dDCADA	1	0	
10	Bob Peterson	Bob	Peterson	M	1/18/1961	USA	dDDbAb	0	0	
19	Allison Janney	Allison	Janney	F	11/19/1959	USA	ACadaC	1	1	
23	Bill Hunter	Bill	Hunter	M	2/27/1940	USA	dCBCdA	0	1	
31	Tom Hanks	Tom	Hanks	M	7/9/1956	USA	bCAcbd	1	1	
32	Robin Wright	Robin	Wright	F	4/8/1966	USA	BdBBAd	1	0	
33	Gary Sinise	Gary	Sinise	M	3/17/1955	USA	DABcbD	1	0	
35	Sally Field	Sally	Field	F	11/6/1946	USA	CAdAAA	2	2	
48	Sean Bean	Sean	Bean	M	4/17/1959	England	cDccBC	0	0	
62	Bruce Willis	Bruce	Willis	M	3/19/1955	Germany	BCDDaa	0	0	
64	Gary Oldman	Gary	Oldman	M	3/21/1958	England	DdAddD	1	0	
65	Ian Holm	Ian	Holm	M	9/12/1931	England	ABdBdD	2	0	
85	Johnny Depp	Johnny	Depp	M	6/9/1963	USA	Cbdcda	1	0	
99	Rob Tipping	Rob	Tipping	M	1/1/1975	Australia	DADdad	3	0	
103	Mark Ruffalo	Mark	Ruffalo	M	11/22/1967	USA	bBCcdD	2	0	
108	Peter Jackson	Peter	Jackson	M	10/31/1961	New Zealand	ADAdBb	0	0	
109	Elijah Wood	Elijah	Wood	M	1/28/1981	USA	AcaaAC	1	3	
110	Viggo Mortensen	Viggo	Mortensen	M	10/20/1958	USA	CACCac	2	1	
112	Cate Blanchett	Cate	Blanchett	F	5/14/1969	USA	BBDCcC	0	1	
113	Christopher Lee	Christopher	Lee	M	5/27/1922	England	acdBCb	1	0	
114	Orlando Bloom	Orlando	Bloom	M	1/13/1977	England	bbcAaA	1	2	
116	Keira Knightley	Keira	Knightley	F	3/26/1985	England	BAABBa	0	1	
131	Jake Gyllenhaal	Jake	Gyllenhaal	M	12/19/1980	USA	DDcDaA	2	1	
134	Jamie Foxx	Jamie	Foxx	M	12/13/1967	USA	acBAcC	2	0	
139	Uma Thurman	Uma	Thurman	F	4/29/1970	USA	DBCAdA	0	0	
140	Lucy Liu	Lucy	Liu	F	12/2/1968	USA	AdDbDa	1	2	
147	Michael Madsen	Michael	Madsen	M	9/25/1957	USA	cBAACd	1	1	

Figure 1: ACTOR Table Datasheet

In terms of Filter, there are several ways to filter but I decide to choose any record in the Gender attributes equal to “M”(Male). Subsequently, in the *Sort & Filter* group in *Home* tab, I choose *Selection* and select “*Equals M*”.

ActorNo	FullName	GivenName	Surname	Gender	Birthdate	BirthCountry	Password	NoOfDogs	NoOfCats
3	Harrison Ford	Harrison	Ford	M	7/13/1942	USA	AbAaAb	3	3
7	Andrew Stanton	Andrew	Stanton	M	12/3/1965	USA	aaabaD	1	3
8	Lee Unkrich	Lee	Unkrich	M	8/8/1967	USA	dDCADA	1	0
10	Bob Peterson	Bob	Peterson	M	1/18/1961	USA	dDDbAb	0	0
23	Bill Hunter	Bill	Hunter	M	2/27/1940	USA	dCBCdA	0	1
31	Tom Hanks	Tom	Hanks	M	7/9/1956	USA	bCAcbd	1	1
33	Gary Sinise	Gary	Sinise	M	3/17/1955	USA	DABcbD	1	0
48	Sean Bean	Sean	Bean	M	4/17/1959	England	cDccBC	0	0
62	Bruce Willis	Bruce	Willis	M	3/19/1955	Germany	BCDDaa	0	0
64	Gary Oldman	Gary	Oldman	M	3/21/1958	England	DdAddD	1	0
65	Ian Holm	Ian	Holm	M	9/12/1931	England	ABdBdD	2	0

Figure 2: Filter field Gender with criteria M

Regarding the query, choose the **Create** tab, click **Query Design**, add the **ACTOR** table, drag all fields to the design grid as shown in the image below, and enter **"M"** in the **Criteria** row of the **Gender** field.

Field:	ActorNo	FullName	Gender
Table:	ACTOR5520	ACTOR5520	ACTOR5520
Sort:			
Show:	☒	☒	☒
Criteria:			"M"
or:			

Figure 3: Query Design showing all actors with Gender = "M"

Query_Gender		
ActorNo	FullName	Gender
3	Harrison Ford	M
7	Andrew Stanton	M
8	Lee Unkrich	M
10	Bob Peterson	M
23	Bill Hunter	M
31	Tom Hanks	M
33	Gary Sinise	M
48	Sean Bean	M
62	Bruce Willis	M
64	Gary Oldman	M
65	Ian Holm	M
85	Johnny Depp	M
99	Rob Tipping	M
103	Mark Ruffalo	M
108	Peter Jackson	M
109	Elijah Wood	M
110	Viggo Mortensen	M
113	Christopher Lee	M
114	Orlando Bloom	M
131	Jake Gyllenhaal	M
134	Jamie Foxx	M
147	Michael Madsen	M
190	Clint Eastwood	M
192	Morgan Freeman	M
193	Gene Hackman	M
194	Richard Harris	M
206	Jim Carrey	M
207	Tom Wilkinson	M
212	David Cross	M

Figure 4: Query Result Showing Male Actors from the ACTOR Table

B. Task Pass1b

After downloading the file moviedatasetDAD.accdb, I renamed the following tables including **ACTOR**, **CASTING**, **COLOURTYPE**, **MOVIE**, **RATING** into **ACTOR5520**, **CASTING5520**, **COLOURTYPE5520**, **MOVIE5520**, and **RATING5520**.

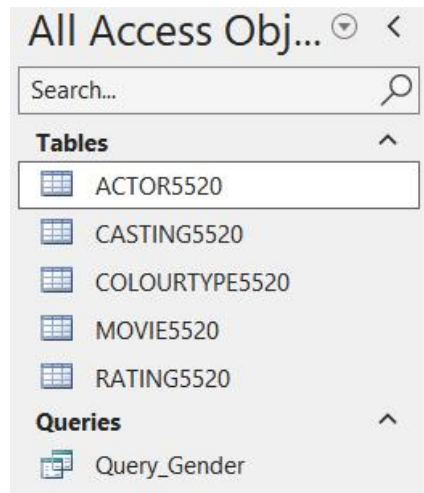


Figure 5: Renamed Database Tables with Student ID Suffix (5520)

Next, according to query description, required field must be contain **MOVIE5520**, **TITLE**, **RATINGCODE**, **RELYEAR** (Must be in 2012), **RUNTIME** (Descending Sort). After finishing query, save query to named **T1PB_5520**.

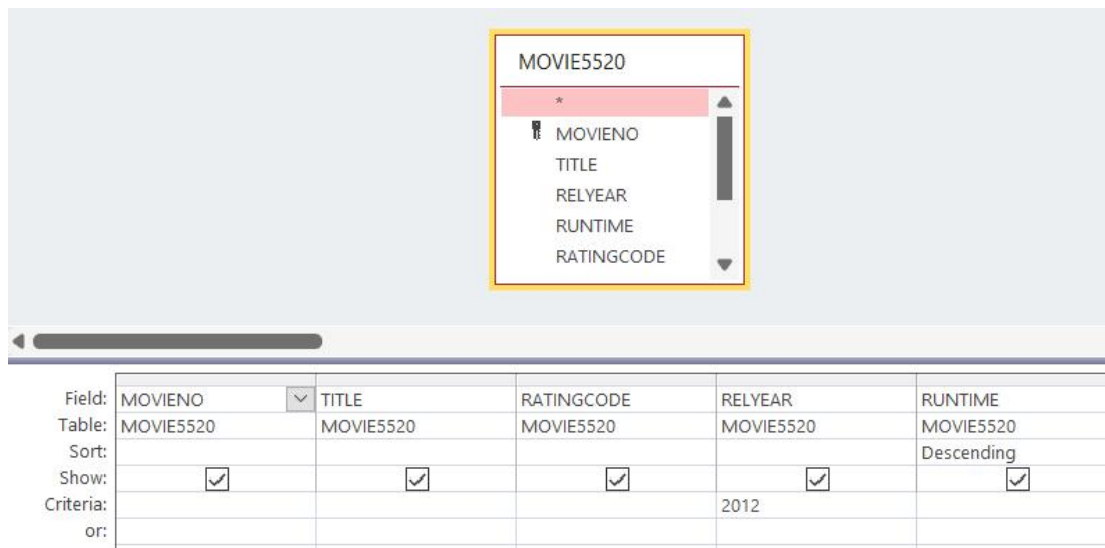


Figure 6: Query Design for Movies Released in 2012 Sorted by Runtime in Descending Order

MOVIEENO	TITLE	RATINGCODE	RELYEAR	RUNTIME
83542	Cloud Atlas	MA	2012	172
49051	The Hobbit: An Unexpected Journey	PG	2012	169
49026	The Dark Knight Rises	M	2012	165
68718	Django Unchained	MA	2012	165
82695	Les Miserables	M	2012	157

Figure 7: Query Result Showing Movies Released in 2012 Sorted by Runtime (Descending)

C. Task Pass1c

Based on the assignment requirement, task Pass1c require a query named **TIPC_5520** which must be display all he MOVIEENO, TITLE RATINGCODE, RELYEAR, RUNTIME, and TMDB_SCORE for all movies that were made in the period 2002-2010 (inclusive), have a runtime less than 120 minutes, are rated "G" or "PG", and have a TMDB score greater than or equal to 6.9. The list must be in ascending Movieno sequence.

Field:	MOVIEENO	TITLE	RATINGCODE	RELYEAR	RUNTIME	TMDB_SCORE
Table:	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520
Sort:	Ascending					
Show:	☒	☒	☒	☒	☒	☒
Criteria:			"G" Or "PG"	Between 2002 And 2010	<120	>=6.9
or:						

Figure 8: Query Design for Filtered Movies (Pass 1c)

MOVIEENO	TITLE	RATINGCODE	RELYEAR	RUNTIME	TMDB_SCORE
118	Charlie and the Chocolate Factory	PG	2005	115	6.9
773	Little Miss Sunshine	PG	2006	102	7.6
1584	School of Rock	PG	2003	110	6.9
7326	Juno	PG	2007	96	7
9502	Kung Fu Panda	PG	2008	90	7
10193	Toy Story 3	G	2010	103	7.7
10625	Mean Girls	PG	2004	97	7.1
11631	Mamma Mia!	PG	2008	108	6.9
22538	Scott Pilgrim vs. the World	PG	2010	112	7.3

Figure 9: Query Result Showing Filtered Movies (Pass 1c)

D. Task Pass1d

According to the requirement of Task1d in the assignment, the query returns all columns from table named **MOVIE5520** with following criteria:

- The movie title should match with this criteria: Like *"*man*" Or Like "*boy*" Or Like "*miss*" Or Like "*girl*" Or Like "*ms*"*.
- The movie must have fewer than 2700 TMDB votes (Criteria of TMDB: *<2700*) and Descending

Field:	MOVIE520	TITLE	RELYEAR	RUNTIME	RATINGCODE	COLOURCODE	TMDB_SCORE	TMDB_VOTES
Table:	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520	MOVIE5520
Sort:								Descending
Show:	☒	☒	☒	☒	☒	☒	☒	☒
Criteria:		or Like "ms"						<2700
or:								

Figure 10: Query Design for Movie Titles Containing Specific Keywords (Pass 1d)

MOVIE520	TITLE	RELYEAR	RUNTIME	Expr1004	RATINGCODE	COLOURCODE	TMDB_SCORE	TMDB_VOTES	IMDB_ID
414	Batman Forever	1995	121 M		M	C	5.3	2551	tt0112462
415	Batman & Robin	1997	125 M		M	C	4.2	2465	tt0118688
8844	Jumanji	1995	104 PG		PG	C	7	2361	tt0113497
8699	Anchorman: The Legend of Ron Burgundy	2004	95 M		M	C	6.8	2214	tt0357413
1493	Miss Congeniality	2000	111 M		M	C	6.4	1776	tt0212346
512200	Jumanji: The Next Level	2019	98 PG		PG	C	7.3	1671	tt7975244
816	Austin Powers: International Man of Mystery	1997	94 M		M	C	6.5	1613	tt0118655
314	Catwoman	2004	104 PG		PG	C	4.4	1570	tt0327554
245	About a Boy	2002	101 M		M	C	6.6	1084	tt0276751
804	Roman Holiday	1953	119 PG		PG	B	7.9	848	tt0046250
8963	Crimson Tide	1995	116 M		M	C	7.1	759	tt0112740
2067	Mission to Mars	2000	114 PG		PG	C	5.9	601	tt0183523
403	Driving Miss Daisy	1989	99 PG		PG	C	7.2	588	tt0097239
398978	The Irishman	2019	209 MA		MA	C	8.6	452	tt1302006

Figure 11: Query Result Showing Movie Titles Containing Specific Keywords (Pass 1d)

E. Task Pass1e

According to description of task1e, the task requires me to generate a table named Member_5520 with following configure:

Field Name	Data Type
Membid (Primary Key)	Number
Surname	Short Text
GivenName	Short Text
Birthdate	Date/Time

Field Name	Data Type
Membid	Number
Surname	Short Text
Givenname	Short Text
Birthdate	Date/Time

Figure 12: Table Design Screen for Member Table

Member_5520				
Membid	Surname	Givenname	Birthdate	Click to Add
1	Bloggs	Fred	5/6/1997	
2	Soe	Davies	4/13/2001	
3	Emma	Pitt	8/23/1992	
4	Tran	Hieu	9/28/1999	
105565520	Truong	Hieu	9/8/2006	
*				

Figure 13: Data Rows in Member Table

F. Task Pass1f

For creating relationship, click on the **Database Tools** and then choose **Relationships**. After that, pull out all five tables in the left side to the center and then create relationships. For instance, connecting the primary key ActorNo in ACTOR5520 table with foreign key ActorNo in CASTING5520 table.

Figure 14: Edit Relationships Dialog Box

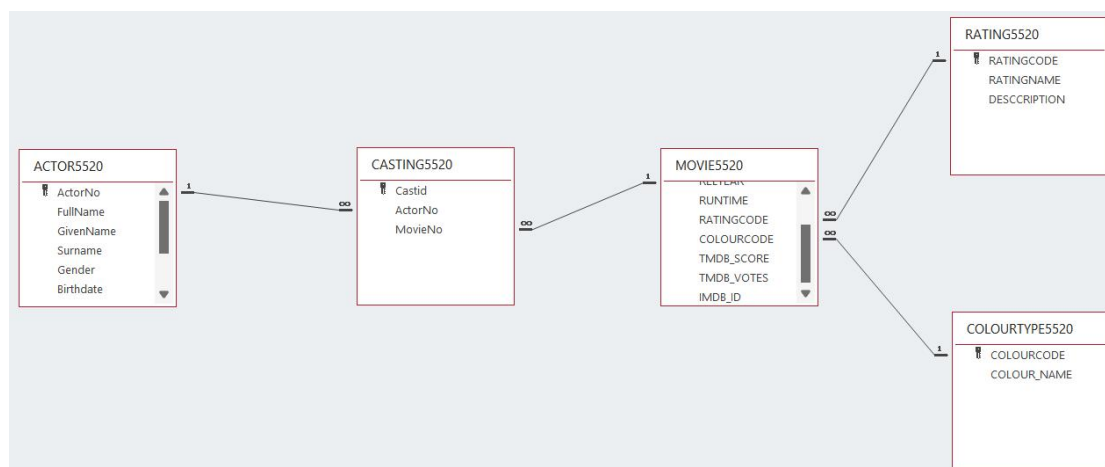


Figure 15: Table Relationships Diagram

G. Task Pass1g

In detail, task Pass1g requires student to query which must include following column (attributes): **MOVIE**NO (from table **MOVIE5520**), **TITLE** (from table **MOVIE5520**), **RATING**CODE (from table **RATING5520**), **DESCRIPTION** (from table **RATING5520**), and **COLOUR**_NAME (fromt table **COLOURTYPE5520**).

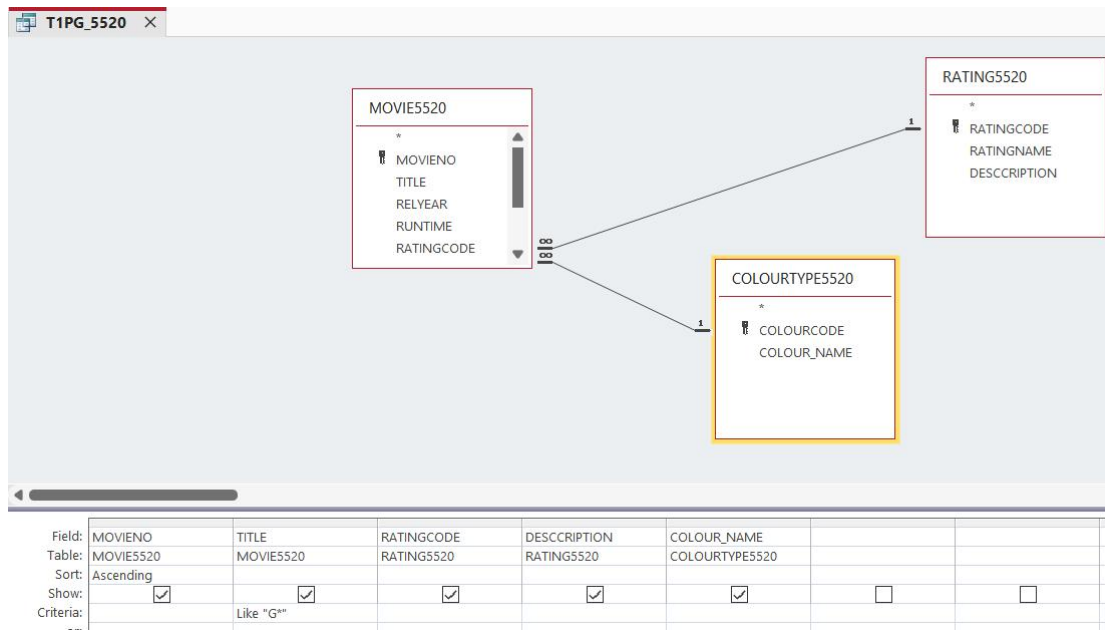


Figure 16: Query Design for Movies Beginning with "G" (Pass 1g)

MOVIEENO	TITLE	RATINGCODE	DESCRIPTION	COLOUR_NAME
38	Gladiator	M	Persons under 15 years must be accompanied by a mature adult	Colour movie
137	Groundhog Day	PG	Recommended for mature audiences 15 years and over	Colour movie
489	Good Will Hunting	M	Persons under 15 years must be accompanied by a mature adult	Colour movie
620	Ghostbusters	PG	Recommended for mature audiences 15 years and over	Colour movie
11520	Grumpy Old Men	PG	Recommended for mature audiences 15 years and over	Colour movie
11665	Get Smart	PG	Recommended for mature audiences 15 years and over	Colour movie
12157	Green Card	PG	Recommended for mature audiences 15 years and over	Colour movie
43074	Ghostbusters	PG	Recommended for mature audiences 15 years and over	Colour movie
45772	Gnomeo & Juliet	G	Suitable for all ages	Colour movie
49047	Gravity	PG	Recommended for mature audiences 15 years and over	Colour movie
118340	Guardians of the Galaxy	PG	Recommended for mature audiences 15 years and over	Colour movie
210577	Gone Girl	MA	Parental guidance recommended for persons under 15 years	Colour movie
283995	Guardians of the Galaxy Vol. 2	PG	Recommended for mature audiences 15 years and over	Colour movie
353070	Going in Style	PG	Recommended for mature audiences 15 years and over	Colour movie
445571	Game Night	M	Persons under 15 years must be accompanied by a mature adult	Colour movie

Figure 17: Query Result Showing Movies Beginning with "G" (Pass 1g)

H. Task Pass1h

In the task Pass1h, the task description asks students to create a query named T1PH_5520 that requires **Totals** with display the actor no, actor fullname, and the count of CastId with Descending and display the first 15 rows.

Field:	ActorNo	FullName	CountOfCastId: CastId
Table:	ACTOR5520	ACTOR5520	CASTING5520
Total:	Group By	Group By	Group By
Sort:			Descending
Show:	☒	☒	☒
Criteria:			
or:			

Figure 18: Query Design with Totals for Actor Movie Count (Pass 1h)

ActorNo ▼	FullName ▼	CountOfCast ▼
2046029	Juliet Blacher	3325
2216110	Jonathan Groff	3324
105565520	Truong Ngoc Gia Hieu	3323
8	Lee Unkrich	3322
205	Kirsten Dunst	3321
2157	Robin Williams	3320
41087	Leslie Mann	3319
88029	Alison Brie	3318
221581	Rebel Wilson	3317
118545	Dakota Johnson	3316
23659	Will Ferrell	3312
88029	Alison Brie	3311
9281	Elizabeth Banks	3310
73457	Chris Pratt	3309
234352	Margot Robbie	3308

Figure 19: Query Result Showing Actor Movie Counts (Pass 1h)

I. Task Pass1i

In the description of the task Pass 1i, it requires grouping by gender and calculating the total number of each gender. Therefore, the query will include two columns: the first is **Gender**, and the second is **CountOfGender**.

Field:	Gender	Gender
Table:	ACTOR5520	ACTOR5520
Total:	Group By	Count
Sort:		
Show:	☒	☒
Criteria:		
or:		

Figure 20: Query Design Grid for Pass 1i (T1Pi_5520)

T1Pi_5520	
Gender	CountOfGender
F	224
M	602

Figure 21: Query Results for Pass 1i (T1Pi_5520)

J. Task Pass1j

In this part, the **Indexed** property of the **Surname** change to **Yes (with duplicate)**.

T1PK_5520								
Castid	MOVIEID	TITLE	RELYEAR	RATINGCODE	ActorNo	FullName	Birthdate	BirthCountry
948	13	Forrest Gump	1994	PG	31	Tom Hanks	7/9/1956	USA
949	13	Forrest Gump	1994	PG	32	Robin Wright	4/8/1966	USA
950	13	Forrest Gump	1994	PG	33	Gary Sinise	3/17/1955	USA
951	13	Forrest Gump	1994	PG	35	Sally Field	11/6/1946	USA
952	13	Forrest Gump	1994	PG	6751	Siobhan Fallon Hogan	5/13/1961	USA
953	13	Forrest Gump	1994	PG	23967	Mary Ellen Trainor	7/8/1950	USA
954	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	85	Johnny Depp	6/9/1963	USA
955	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	114	Orlando Bloom	1/13/1977	England
956	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	116	Keira Knightley	3/26/1985	England
957	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	378	Jonathan Pryce	6/1/1947	Unknown
958	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	8691	Zoe Saldana	6/19/1978	USA
959	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	4030	Martin Klebba	6/23/1969	USA
960	22	Pirates of the Caribbean: The Curse of the Black Pearl	2003	PG	4031	Greg Ellis	3/21/1968	England

Figure 27: Query Results for T1PK_5520 (Casting, Movie, and Actor Information)